



Smart Order Routing

You are working as a summer intern on the algorithmic trading desk at JPLonegan Chase, one of the world's most prestigious financial institutions. Your company has to work orders for large buy-side institutions in a new fragmented market environment. It is essential to meet regulatory requirements to fill orders at the NBBO (National Best Bid and Offer). In addition, given fragmented markets, managing liquidity risk by minimizing potential price impact has become even more challenging than it used to be when liquidity was focused on the traditional exchanges.

Your task in this ALG03 case:

Your supervisor would like to challenge your skills and has asked you to build a smart order router algorithm for one of the stocks for which the desk deals. Since she knows that you might not be familiar with this concept, she has provided you with a brief description of how smart order routing works. Your supervisor's email is provided below.

Dear intern,

Given the large number of exchanges that are available, it is common to find the same stock traded on several exchanges. For example, Royal Bank of Canada trades in Toronto on the TSX and in New York on the NYSE (as well as on other smaller exchanges). It is also possible for the same stock to trade on two or more exchanges within the same country. In Canada, Royal Bank of Canada trades on both the TSX (the primary and oldest exchange) as well as on the Alpha Trading System (commonly known as Alpha). The latter is an alternative equity marketplace that competes with the main exchanges. For this case, we will assume that take fees are \$0.01 and \$0.005 on the main and alternative exchanges respectively; and that you will receive a rebate of \$0.0075 per share for limit orders that are filled on the alternative exchange.

Smart Order Routing (SOR) is a mechanism by which an algorithm discovers the best price and available liquidity for a given security across all relevant exchanges. The algorithm will determine to which exchanges orders have to be sent in order to obtain the best price. This requires monitoring of both quotes and associated volumes since price fills will reflect any price impact due to limited liquidity at the quoted prices. The algorithm will break large orders into smaller orders and route them in a way that manages that liquidity risk.

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An example will help you understand how a smart order router works. Consider the following order books for stock THOR in a Main Market and an Alternative Market, respectively:

Main Market			
Bid Volume	Bid Price	Ask Price	Ask Volume
6,700	19.97	19.98	7,400
5,400	19.96	19.99	5,400
5,500	19.95	20.00	5,600
5,300	19.94	20.01	5,300
6,000	19.93	20.02	5,300
4,800	19.92	20.03	6,300

Alternative Market			
Bid Volume	Bid Price	Ask Price	Ask Volume
9,000	19.97	19.98	9,500
8,300	19.95	20.00	9,700
8,400	19.93	20.02	9,000
8,000	19.91	20.04	7,200
8,000	19.89	20.06	8,100
9,000	19.87	20.08	8,500

Consider the case where one of our clients would like to buy 100,000 shares of THOR from our bank. After a brief negotiation with our colleagues in Sales, the client agrees that he is willing to buy the shares at a premium and pay \$20.05/share. Since our colleagues in Sales do not have a model to evaluate whether \$20.05/share is a good price for THOR given the available liquidity, they will ask for your input and you will have to decide whether to approve the transaction.

In order to make a decision, you need to evaluate the liquidity of the two markets together and in order to do so you should build the “Aggregate Order Book”, a limit order book that combines all of the orders from both the Main and the Alternative Exchanges. The aggregate order book for THOR is provided below based on the quotes of the Main and Alternative markets. For ease of presentation, we will only show the ask side of the book here but the same calculations can be done for the bid side.

Aggregate Order Book			
Ask Price	Ask Volume	Cumulative Ask Volume	VWAP
19.98	16,900	16,900	19.980
19.99	5,400	22,300	19.982
20.00	15,300	37,600	19.990
20.01	5,300	42,900	19.992

20.02	14,300	57,200	19.999
20.03	6,300	63,500	20.002
20.04	7,200	70,700	20.006
20.06	8,100	78,800	20.012
20.08	8,500	87,300	20.018

When evaluating the order, you can now make a decision and have more details about the liquidity for THOR stock. If you decide to accept the transaction with the client as proposed by your colleagues in Sales, our bank will sell 100,000 shares at 20.05. Since we do not have any estimates for the future THOR stock price, we are required to hedge against market price risk¹; therefore, we need to buy back shares. In this example, you will notice that you can buy 70,700 shares at a price lower than \$20.05 (as shown in the Cumulative Ask Volume column). You might accept some risks and decide to take the order; you will be able to cover most of your exposure to market price risk using market orders and then work the remaining part of the order using limit orders to minimize liquidity risk.

The example above is easier than most of the offers that our bank faces on a daily basis. Therefore, I would like you to build a decision-support spreadsheet with RTD links to the simulated markets in order to collect the relevant information about quotes and liquidity in real time. Using that information, you will then need to program an algorithm that covers accepted orders in a way that manages both market price risk and liquidity risk.

As described in the email above by your supervisor, you will need to use your knowledge of liquidity risk and market price risk to evaluate the riskiness of the order, accept or decline it depending on the tradeoff between risk and potential returns, and then use your algorithm to cover your position appropriately. To evaluate the tradeoff between risk and potential returns, it will be useful to review some measures of liquidity.

Review of some measures of liquidity:

Recall that there are several related measures of liquidity or illiquidity associated with a security trading on a pure order-driven market. For example:

- Bid-Ask spread
- Breadth and Depth of the limit order book (volume-based measures)
- Resiliency
- Price impact

Which measure(s) one uses to quantify liquidity risk will depend on one's role and consequently on one's exposure to the costs of illiquidity. The bid-ask spread measure is very important for a market

¹ For this case, we define market price risk as the risk that the market price of the security will change over time for exogenous reasons; whereas the implication of illiquidity will involve potential price impact associated with your order (endogenous price changes).

maker (please see the ALGO2 case for more details) but it is less important in this ALGO3 case. This is because, as in the LT3 and LT4 cases, you will only be looking at one side of the market while ‘working’ an order since you will only be buying or selling depending on the trade that you executed with their client.

Breadth and depth of the limit order books measure volume available to trade. Depth of the market measures the number of shares that are available to buy (or sell) at a particular price. It is an indicator of how large the order has to be before causing a price change (moving down the book). In the example below, the depth of the market at the first level of the book’s bid quotes is 1,000.

Bid Volume	Bid Price	Ask Price	Ask Volume
1000	9.98	10.00	5000
4000	9.95	10.03	3000
4500	9.90	10.07	2000
10000	9.88	10.42	18000
15000	9.84	10.45	25000

Breadth is how many levels of orders (at different price quotes as one moves down the book) are currently available in the book. That is, breadth of the limit order book indicates whether the orders are numerous across a wide range of price levels. Breadth and depth of the market are strongly correlated, as markets with good depth also tend to foster breadth.

One useful way to measure liquidity from the perspective of a liability trader is to evaluate how much the price would change if a large market order were to be submitted. The less the price change for a specific volume, the more liquid the market is. This is called price impact and has been discussed in the Liability Trading 3 case (LT3). The Excel decision-support spreadsheets for the LT3 case and the multiple exchange LT4 case compute potential price impact for any market order size in real time. The LT4 version provides a global limit order book for which liquidity available in all of the exchanges is aggregated and combined into one limit order book. This information can be used to compute the potential price impact associated with the optimal routing of the order across the available exchanges.

Another measure that can be used to evaluate the liquidity is the “resiliency”. Resiliency can be measured by the re-fill rate, that is, how quickly new limit orders are submitted to the visible book when liquidity is removed from the book. Resiliency measures the time for the price to recover after a large trade. It represents the degree to which price movements are smooth. Several ratios have been developed by both academics and practitioners to measure the resiliency in the market. One that can be easily applied in the RIT is the Market Efficiency Coefficient (MEC) that measures differences between the volatility in a short-period compared to the volatility over a long-period. More formally, let $var(r_{weekly})$ be the weekly variance of the returns and $var(r_{daily})$ be the daily variance of the returns, assuming that there are 5 trading days in a week, the MEC is calculated as:

$$MEC = \frac{var(r_{weekly})}{[5 * var(r_{daily})]}$$

When MEC is lower than 1, it suggests that the market is illiquid because the variance of the short-period returns is higher than the variance of the long-period returns. In other words, this ratio provides some information on the proportion between temporary and longer-term price movements.

You can easily implement this ratio in the RIT. For example, use a 30 second moving window for the long-period and a 6 second moving window for the short-period. Link the historical prices to Excel and do the calculations to come up with a real-time MEC².

Execution

If you wanted to buy 20,000 shares of THOR, how could you do it? Should you trade on only one market? The objective of the case is to process the orders and seek the best execution. Therefore, you should take advantage of liquidity on both exchanges. In the example above, we could calculate the aggregate order book as shown below – again, for ease of presentation, we will only show the ask side of the book since our example requires you to buy 20,000 shares; but the same calculations can be done for the bid side.

Aggregate Order Book				
Ask Price	Ask Volume	Cumulative Ask Volume	VWAP	Market
19.98	7,400	7,400	19.980	Main
19.98	9,500	16,900	19.980	Alternative
19.99	5,400	22,300	19.982	Main
20.00	5,600	27,900	19.986	Main
20.00	9,700	37,600	19.990	Alternative
20.01	5,300	42,900	19.992	Main
20.02	5,300	48,200	19.995	Main
20.02	9,000	57,200	19.999	Alternative
20.03	6,300	63,500	20.002	Main
20.04	7,200	70,700	20.006	Alternative
20.06	8,100	78,800	20.012	Alternative
20.08	8,500	87,300	20.018	Alternative

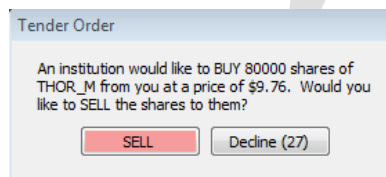
If you would like to buy 20,000 shares using market orders, you should submit orders on both markets to obtain the best price. In particular, you should submit an order for 10,500 shares on the Main market and an order for 9,500 shares on the Alternative market. By doing so, you would buy 7,400 shares at \$19.98 from the Main market, 9,500 shares at \$19.98 from the Alternative market and 3,100 shares at \$19.99 from the Main market. The Volume Weighted Average Price (VWAP) of the transaction will be \$19.982.

² For more information on how to link the real-time data from the RIT in Excel, please check our “RIT – User Guide – RTD Documentation” file available in the RIT Client under “Help Files”.

If you traded the 20,000 shares only on the Main market, the VWAP of the transaction would have been ~\$19.991 and you would have caused the price to move to \$20.01. If you traded the 20,000 shares only on the Alternative market, the VWAP of the transaction would have been ~\$19.991 and you would have caused the price to move to \$20.02. Trading only on one of the exchanges would have resulted in a higher VWAP.

Market Dynamic

During this ALGO3 case, you will receive 5 institutional orders. The institutional orders will be shown as pop-up messages like the one shown below. You will have 30 seconds to decide whether to accept the offer or not. You can track the time remaining before the offer expires by checking the number between brackets beside the “Decline” button.



The simulation will last 5 minutes. There is only one stock trading in this case, THOR. You have been given a net trading limit of 100,000 shares and a gross trading limit of 250,000 shares. There is a maximum order size of 25,000 shares. Take fees are \$0.01 and \$0.005 on the main and alternative exchanges respectively; and that you will receive a net rebate of \$0.0025 per share for limit orders that are filled on the alternative exchange. Marketable limit orders (limit orders that cross the bid/ask spread) are treated as market orders.

RTD Links and API will be enabled allowing you to link the RIT with Excel and VBA. You are allowed to submit orders only through your algorithm and watch the order book for the different exchanges in Excel (the order books are not available in the RIT). Trading from the RIT will be disabled.

Lastly, as in our LT3 and LT3 Decision Cases, your job as a liability trader is to only trade to unwind positions that you accumulated as a consequence of transactions with institutional clients (accepted tender offers). You should not trade for any other reason (e.g., arbitrage, market making, following the trend, etc.). Your instructor or CRO will penalize you for front-running and other speculative strategies, defined for this case as trading unrelated to unwinding accepted tender offers and not completely covering accepted tenders prior to the end of the trading period.

Discussion Question:

- 1) How can you use the breadth and depth of the book to help evaluate the institutional orders?

Instructors' Notes

The Liability Trading (LT) Case sequence challenges participants to develop models in order to evaluate and quantify the risks and the potential returns associated with accepting and covering tender offers (block trades). The ALGO3 case extends the LT4 simulation by introducing a smart-order router (SOR) as an algorithm to manage tender offers across two alternative exchanges.

As stated in the LT case briefs, participants should only trade to *unwind* positions accumulated from transactions with institutional clients (accepted tender offers). To provide an incentive for students to focus on the case objectives and to identify and acquire the associated skills, instructors have the option to fine students for front-running and other speculative strategies, defined for this case as trading unrelated to unwinding accepted tender offers and any nonzero positions at the end of the trading period.

The 'LIABILITY_RISK' case fines module (in the 'Modules' sheet in the server casefile) includes four parameters. Parameter 1 sets the \$ per share fine rate associated with front-running and speculative strategies up to a threshold of the number of shares set by Parameter 2. For shares in excess of that threshold, the fine rate is increased to that specified as Parameter 3. Finally, Parameter 4 sets the fine rate for shares not covered by the end of the case iteration. Note that the threshold number of shares (Parameter 2) applies to the entire iteration and aggregated for all stocks.

For the ALGO3 case, the values for these parameters are set at \$0.20, 5000, \$0.40 and \$0.50 in the 'Modules' sheet of the server casefile. Of course, instructors can choose to change the fine rates or the threshold # of shares. Note that since the same security trades on two exchanges, *BOTH* tickers are fined at the end of the case which is why the fine rate for uncovered shares at the end of the iteration is set at $\frac{1}{2}$ that of the fine rate for uncovered shares at close of trading in single exchange cases such as the LT3 case.

Students will be able to observe any case fines in their transaction log. In their performance report (pushed out at the end of each iteration), they will also see their aggregated fines for each security in the 'Case Fines Summary', and the 'Security Orders Summary'. The 'Trader P&L Chart' plots their net of fines P&L and the P&L (No Fines). Finally, they will see their 'Total Fines' (which includes case fines as well as any trading limit fines) in the 'Trader Info' screen, on their RIT User Application interface.